This booklet is part of a continuing patient education series on the thyroid gland and thyroid disease. Additional booklets available in this series are:

Getting to Know Your Medication
How the Thyroid Gland Can Affect Your Weight
Hyperthyroidism and Your Health
Hypothyroidism and Your Health
The Postpartum Period and Your Thyroid
Thyroid Disease and Aging
The Thyroid Gland and Your Health
Thyroid Disease and Diabetes
Thyroid Disease in Women
Understanding Goiter
Understanding Graves' Disease
Understanding Thyroid Cancer
Understanding Thyroid Nodules
Understanding Your Child's Thyroid
What Is Hashimoto's Disease?

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Knoll Pharmaceutical Company
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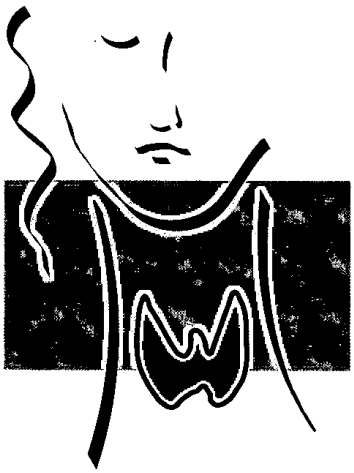
SYN-0297-065/7-97

Printed in USA
EDP 9990193

Understanding TSH Testing



TSH Testing — What You Need To Know

There are a number of tests available for diagnosing thyroid disorders. Currently, the TSH (thyroid-stimulating hormone) test is the most important single test because of its sensitivity to small changes in TSH levels and its ability to detect thyroid disorders at early stages.

The following pages contain basic information about the thyroid gland, its role in your body, and thyroid disorders. In addition, this booklet will discuss the importance of the TSH test in detecting thyroid disorders.

Remember, no single booklet can answer all of your questions or replace speaking with doctors, nurses, and other members of the healthcare team. If you have any questions or concerns about the thyroid or thyroid disorders or the TSH test, please check with your doctor.

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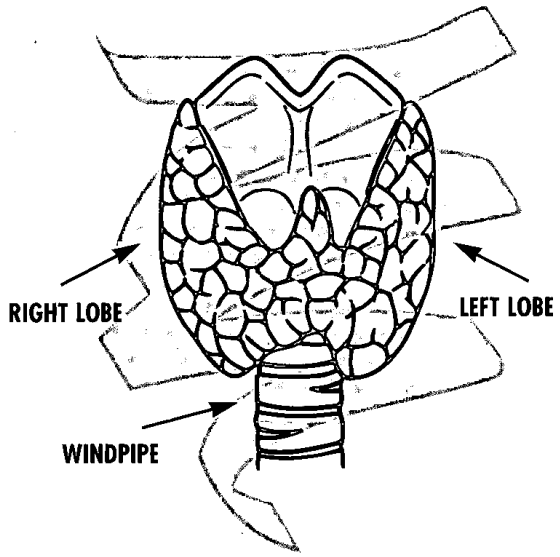
Your Thyroid— The Basics

Your thyroid is a gland located at the base of your neck, just below your Adam's apple. It is shaped like a butterfly—each "wing," or **lobe**, of your thyroid lies on either side of your windpipe.



The purpose of your thyroid gland is to make, store, and release **thyroid hormone** into your blood. Thyroid hormone affects essentially every cell in your body, and helps control your body's functions. If you have too little thyroid hormone in your blood, your body slows down. This is a condition called **hypothyroidism**. If you have too much thyroid hormone in your blood, your body speeds up. This is a condition called **hyperthyroidism**.

For more information on the thyroid gland, please contact:

The Thyroid Foundation of America
Ruth Sleeper Hall, RSL 350
40 Parkman Street
Boston, MA 02114-2698
(send self-addressed, stamped envelope)

The Thyroid Society for Education & Research
7515 S. Main Street, Suite 545
Houston, TX 77030
1-800-THYROID

National Graves' Disease Foundation
2 Tsitsi Court
Brevard, NC 28712

Or visit our web site at:
www.glandcentral.com

A Final Note

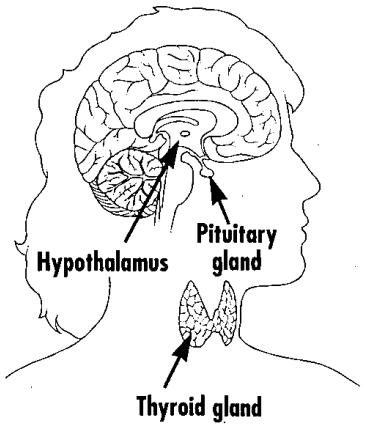
If you believe that you may be suffering from a thyroid disorder, ask your doctor to consider ordering a TSH test. If you are already being treated for a thyroid disorder, your blood TSH level should be checked periodically, at least once a year, to ensure that your treatment continues to be successful.

The amount of thyroid hormone made by your thyroid gland is adjusted by another gland called the **pituitary**, as well as a part of your brain called the **hypothalamus**.

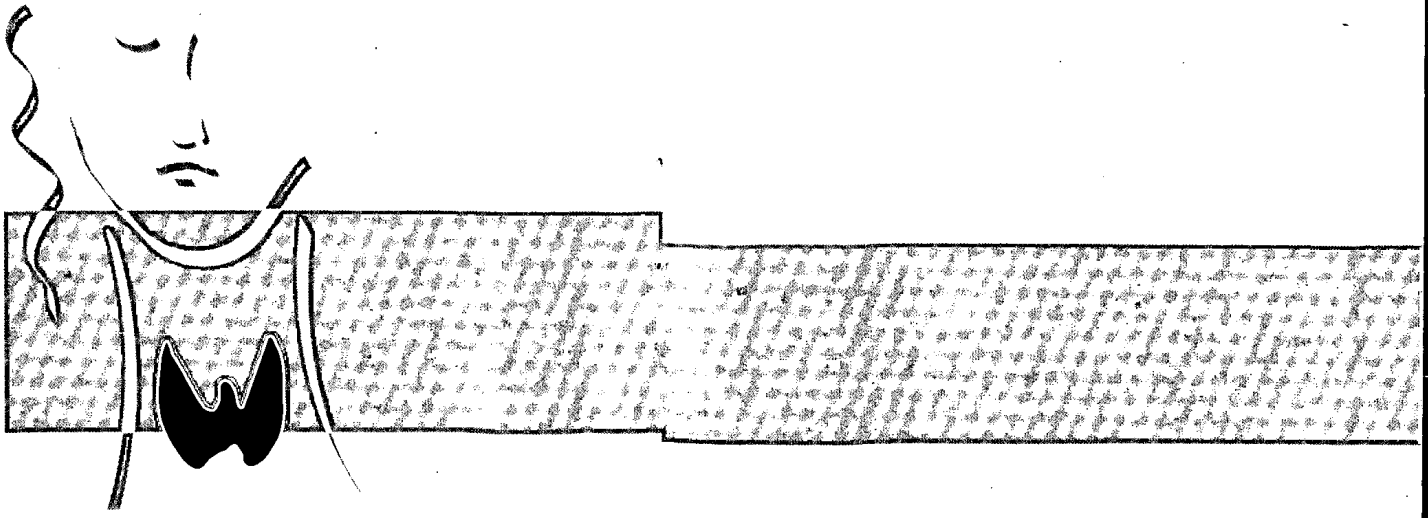
- The pituitary gland makes hormones that control many other glands in your body

- The hypothalamus helps the pituitary control many of your glands, as well as helping to control other bodily functions, such as thirst, hunger, sleep, and body temperature

The thyroid gland, pituitary gland, and hypothalamus all work together to control the amount of thyroid hormone in your body. With the pituitary controlling most of the action, these structures work in a way that is similar to the way a thermostat controls the temperature in a room.



For instance, just as the thermometer in the thermostat senses the temperature of a room, your **pituitary** constantly senses the amount of thyroid hormone in your blood. If there is not enough thyroid hormone, your pituitary senses the need to "turn on the heat." It does this by releasing more **Thyroid-Stimulating Hormone (TSH)**, which signals your thyroid to make more thyroid hormone. Your thyroid gland then makes and releases the hormone directly into your bloodstream and restores the amount of thyroid hormone in your blood to normal.



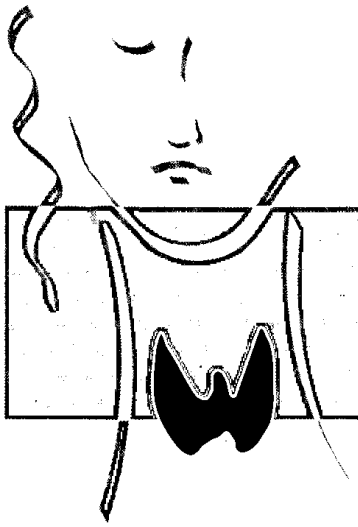
When Is the TSH Test Used?

The TSH test can be used to help determine whether you have a thyroid disorder. Your doctor may choose to measure your thyroid hormone levels as well, to learn more about the nature of the disorder.

The TSH test can also be used to monitor your thyroid function once you have begun treatment for your thyroid disorder. It is especially useful in the treatment of hypothyroidism. Because thyroid hormone replacement therapy is usually needed for life, and the body's specific need for thyroid hormone changes as you age, the TSH test is the most important single method of monitoring thyroid function for any changes that may have occurred.

Just What Is Normal?

TSH level is measured in microunits of TSH per milliliter of blood (μU/mL). Although the normal TSH range can vary somewhat from lab to lab, it is generally considered to include the range of 0.3 μU/mL to 5.0 μU/mL. Therefore, a TSH level lower than 0.3 μU/mL may indicate hyperthyroidism, and a TSH level above 5.0 μU/mL may indicate hypothyroidism.

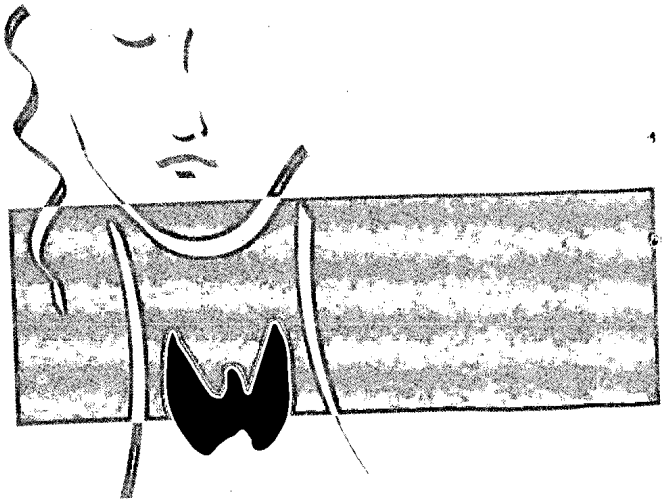


Your pituitary then senses that there is just the right amount of thyroid hormone in your body. With your thyroid hormone levels now restored to normal, your pituitary slows its production of TSH back down to normal.

Thyroid Disorders— A Brief Look at What Can Go Wrong

Hypothyroidism

If your thyroid gland produces too little thyroid hormone, you may have a condition called **hypothyroidism**. If you are hypothyroid, you also have a high level of TSH (which is your body's signal to "turn on the heat," or to step up the production and release of thyroid hormone). Having too little thyroid hormone causes your body to slow down. As a result, you may feel tired, have a slow heartbeat, or feel depressed, besides having other symptoms.



What Can the TSH Test Reveal?

The TSH test is a simple blood test that measures the amount of thyroid-stimulating hormone in your blood. This is important because TSH is your pituitary gland's signal to your thyroid gland to make less or more thyroid hormone.

For example, if you have a high level of TSH in your blood, it usually means that your pituitary senses that there is not enough thyroid hormone in your blood and it is trying to get the thyroid gland to make more.

If you have a low level of TSH, it usually means that your pituitary senses that there is too much thyroid hormone in your blood, and it has decreased its TSH signal.

What Tests Are Available?

There are a number of tests available to help with the diagnosis and treatment of thyroid disorders. Included are tests that measure **thyroid hormone levels** in your blood, as well as the **TSH test**, which measures the concentration of thyroid-stimulating hormone (TSH) in your blood.

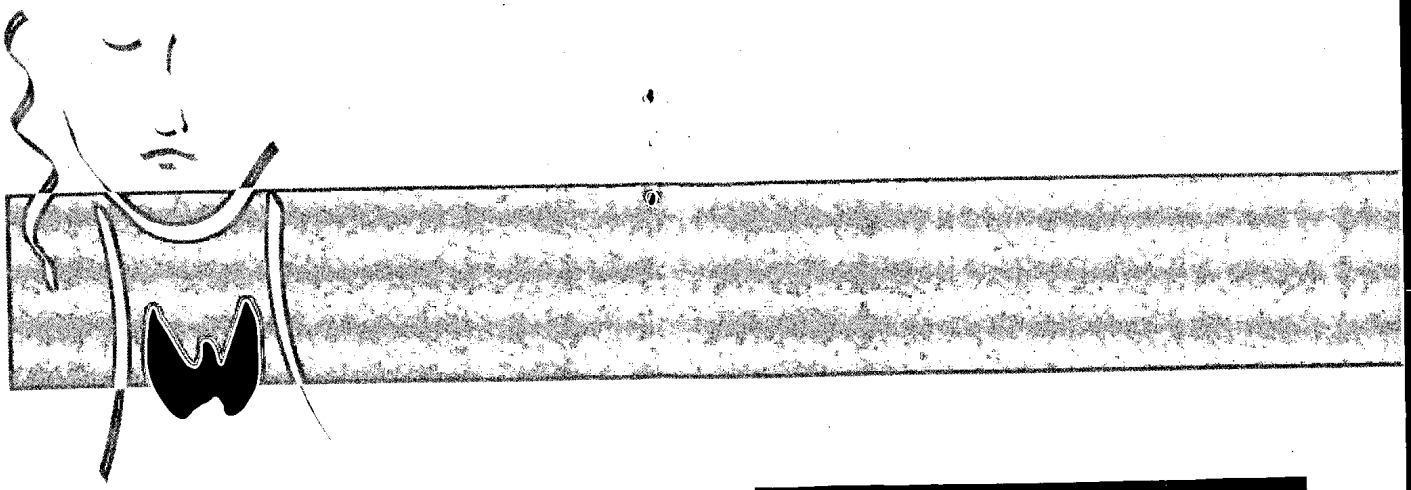
You might wonder why doctors would measure anything other than the amount of thyroid hormone, since that is what needs to be controlled. The reason is that by measuring TSH level, the doctor can detect even the earliest signs of a thyroid disorder.

Mild Thyroid Failure

A more mild form of hypothyroidism is **mild thyroid failure**. In this case, your thyroid hormone levels are normal, but you have a high TSH level. This may mean that TSH is making your thyroid gland work extra hard just to maintain normal levels of hormone in your blood. This condition often leads to hypothyroidism. People with mild thyroid failure generally don't show any obvious symptoms, which makes it very difficult for doctors to identify without performing a TSH test.

Hyperthyroidism

If your thyroid gland produces too much thyroid hormone, you may have a condition known as **hyperthyroidism**. If you are hyperthyroid, you also have a low level of TSH in your blood because your pituitary has decreased its TSH signals to your thyroid gland. Having too much thyroid hormone causes your body to speed up, resulting in nervousness, fast heartbeat, and unexplained weight loss, among other symptoms.

Mild (Subclinical) Hyperthyroidism

If your thyroid hormone levels are normal, but you have a low TSH level, you may have a condition known as **mild (subclinical) hyperthyroidism**. This often progresses to hyperthyroidism at a later point, and should be monitored carefully by your doctor.